

Kommodity

One binary to rule them all.
A sovereign story.

Agenda

KOMMODITY

1. `whoami`
2. `id corti`
3. `man motivation`
4. `kommodity.target`
5. `history | grep commodity`
6. `./kommodity`
7. `grep -r roadmap`

\$ whoami

\$ whoami

Nicklas Frahm
Director of Platform
Engineering



Steffen Karlsson
Principal Platform Engineer



Paul Thuriot
Senior Platform Engineer



1

Mechatronics Engineer

MSc in Mechatronics Engineering with a specialization in Embedded Systems Engineering.

2

Hardware Nerd

Contributor of the NanoPi R5S board overlay to Talos Linux.

3

Nature Lover

When not inside a Kubernetes cluster or a terminal, you will find me in the forest getting bitten by a tick or hiking any available mountain, or rather hill in Denmark.

\$ id corti

The only **Frontier AI Lab** built for clinical reality.

We are about **110 passionate people** building for healthcare developers across the globe.



Medical-grade **speech recognition**

Fast, accurate, and ready to handle clinical conversations.



Purpose-built **text generation**

Turn transcripts, facts, or structured data into any type of medical documentation.



Agentic framework for healthcare

Use clinical-grade models to build any use-case.



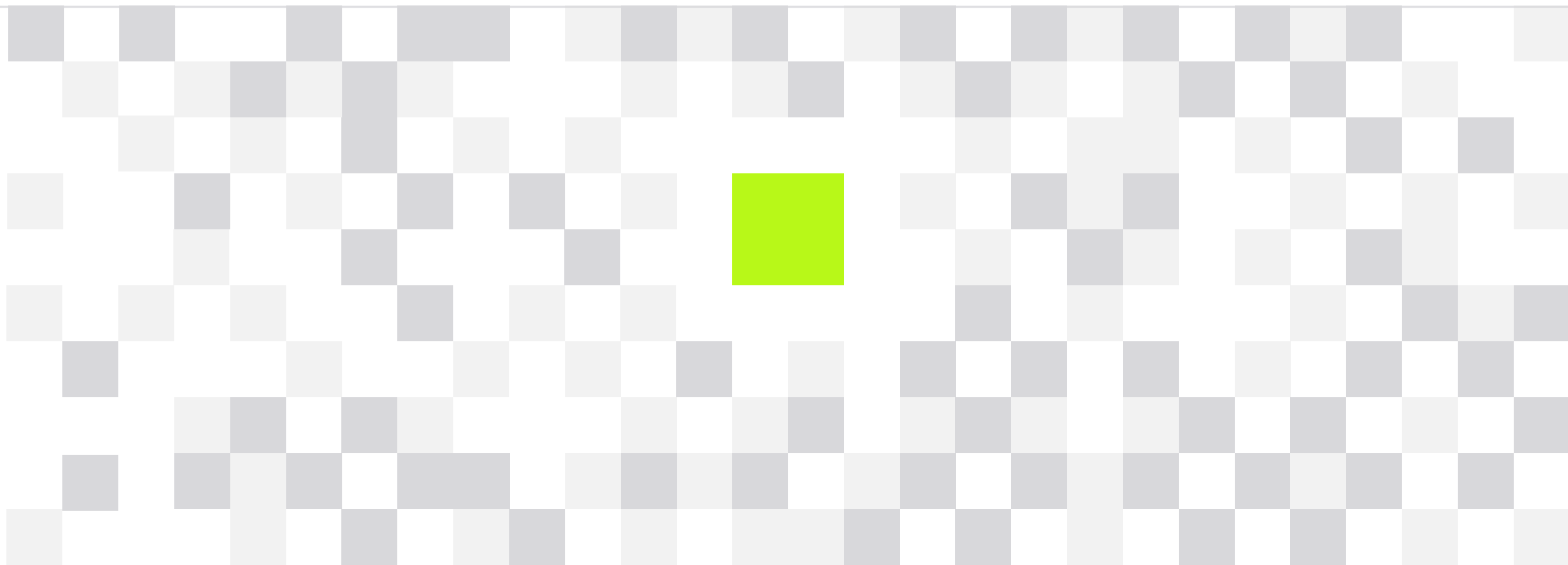
World's most accurate **medical coding**

Based on a comparison with other AI labs.

150,000

patient interviews will be conducted
by a physician over a career.
All happening alone.

*JOUKES E ET AL, 2018



35%

of physicians' time is spent on administration & documentation to minimize this risk.

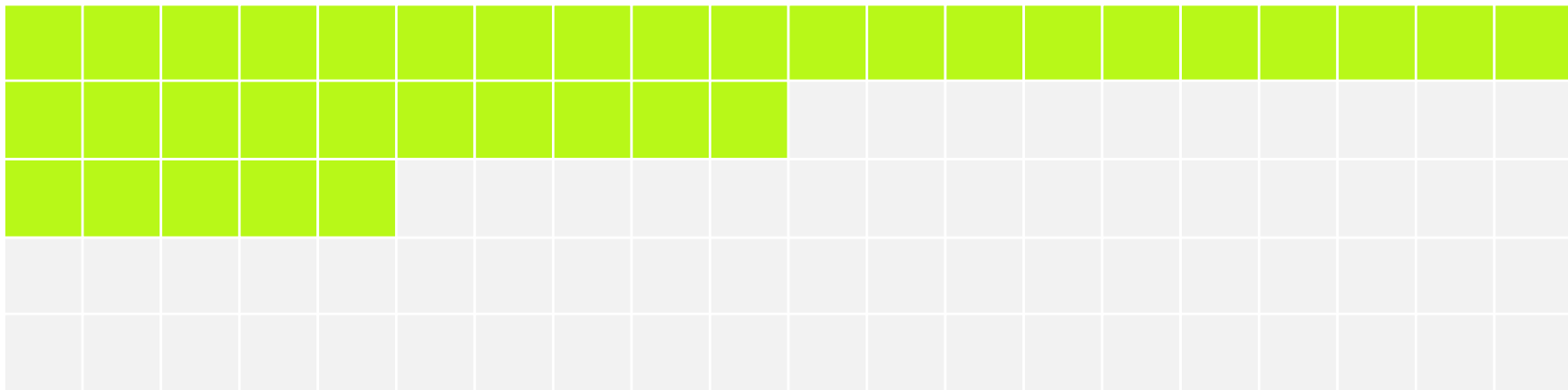
*JOUKES E ET AL, 2018



ADMINISTRATION
& DOCUMENTATION



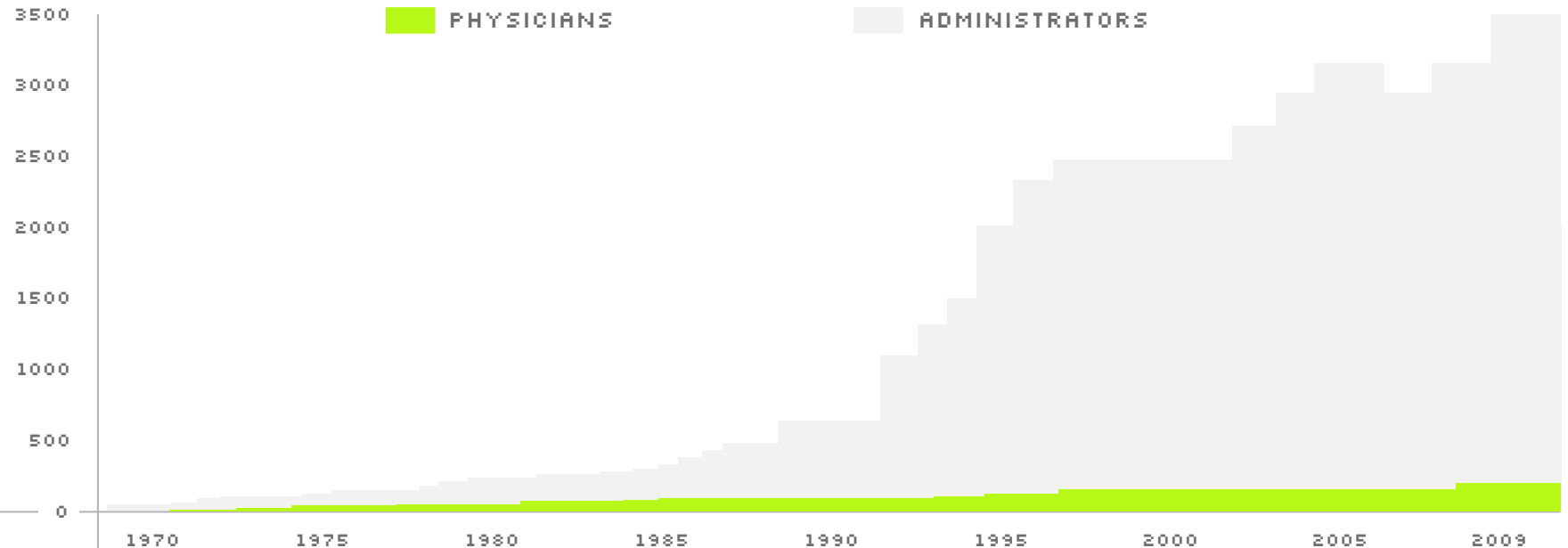
CLINICAL WORK



3,500%

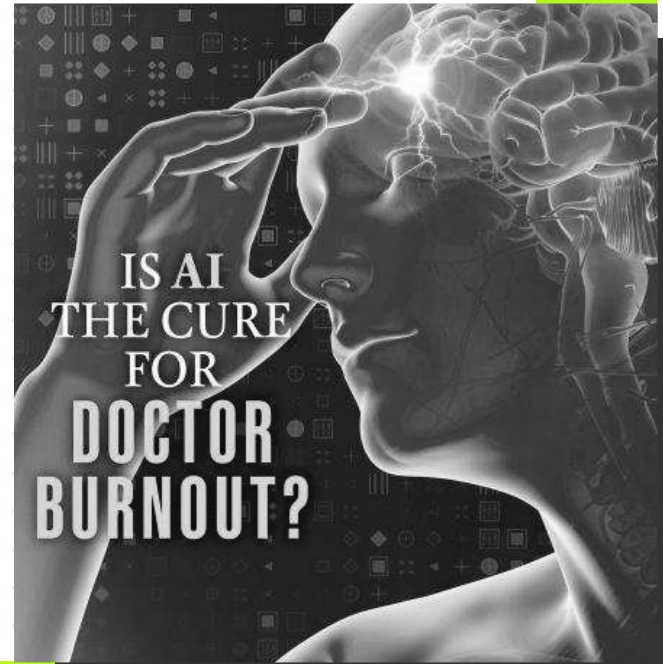
that's how much growth there's been in administrative personnel in healthcare.

*JOUKES E ET AL, 2018



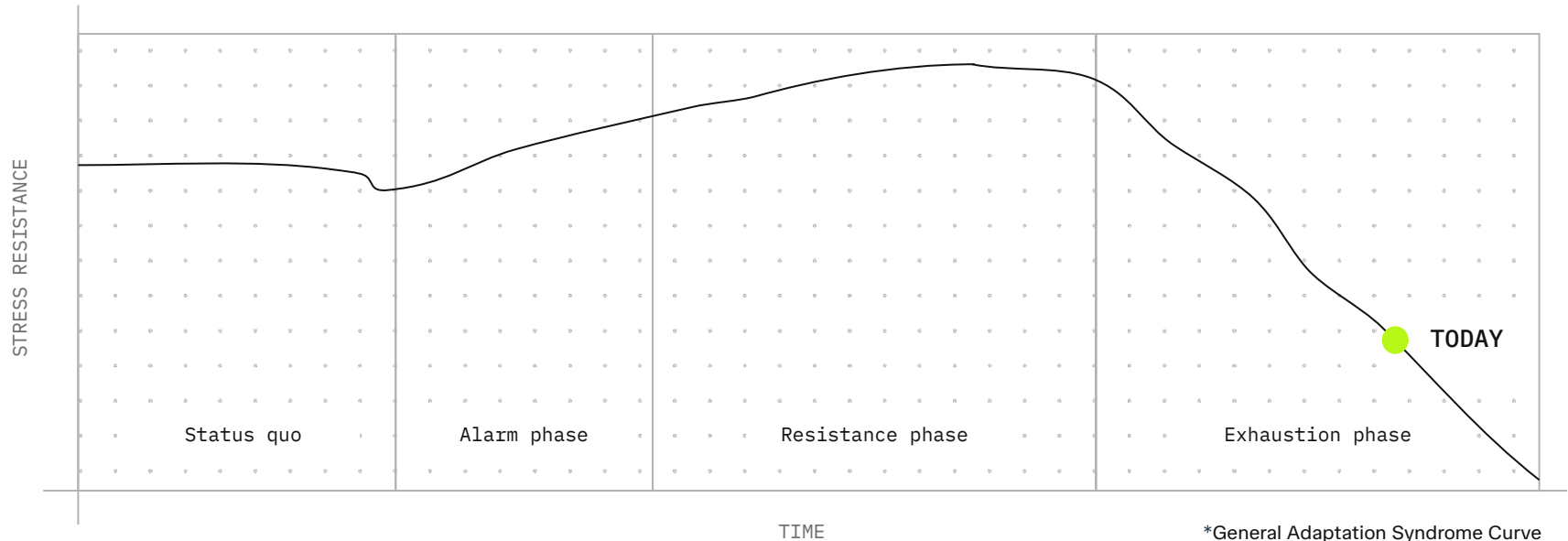
Newsweek asked:

Is AI the cure for
doctor burnout?



THE PROBLEM:

Doctors are tired of change, they are trusting us to get AI right.



*General Adaptation Syndrome Curve

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\$ man motivation

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Why would you build your own **Kubernetes engine**?

What do the haters say?

- > **Scope:** Building a Kubernetes Engine is a huge project!
- > **Skill:** You are not a cloud provider or hyperscaler!
- > **Focus:** This is not your core business!
- > **Solutionism:** Just use OpenShift!

1

How do we get the **best prices** for GPUs?

While partner with Microsoft, we are also aware of capacity and pricing across different providers. We want to serve healthcare and the public sector, which requires competitive pricing.

2

How can we deploy to **Switzerland** and **multi cloud** while keeping our sanity?

Kommodity can build clusters in the cloud or in a datacenter in the Swiss Alps. The management stack is lightweight and only has a dependency on PostgreSQL.

3

How do we **own our tech stack** during times of **geopolitical uncertainty**?

Customers demand workloads to be hosted in Europe, but most recently they also want us to host our infrastructure on a European provider where the Cloud Act does not apply. One of these strategic partners is Scaleway.

4

How do we optimize for **operational simplicity**?

While Kubernetes generally tries to promise portability there are some serious caveats when trying to do this at scale across providers. Instead of reinventing our stack on every provider with their managed Kubernetes offering we chose to rely on the same OS and Kubernetes across providers.

kommodity.target

kommodity.target

Kommodity is an open-source Kubernetes engine built on top of Talos Linux and CAPI.

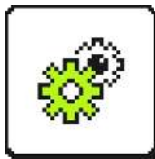
You should own your infrastructure. Network, storage and compute should be a commodity, not bespoke and vendor-locked.

Kommodity makes that a reality by solving network and compute. In combination with storage technologies, such as rook-ceph, CNPG and other Kubernetes operators, Kommodity can fulfill all your infrastructure needs.



Built on top of Talos Linux.

Talos Linux allows us to build secure, immutable and fast infrastructure so we can spin up GPUs in minutes.



Built on top of Cluster API.

Cluster API providers and cloud controller managers as drivers for infrastructure. New drivers can be added in days, instead of months.



A single binary. A single port.

A single static Go binary that only depends on a PostgreSQL database for storage. Compose, Kubernetes, Azure Container Apps, systemd.



Secure and compliant by design.

Private networking, disk encryption, immutable OS and no SSH enable compliance with frameworks such as SOC2, C5, HDS and ISO27001.



Open source. Sovereign requires builders.

Many European companies demand sovereign solutions, but Europe has for a long time mostly used open source rather than contributed.



**\$ history | grep
kommodity**

One does not simply build a single binary.

1

Exposing KMS and Kube API via a single port.

To enable a single port, we use a small library called cmux that identifies gRPC traffic.

2

Authenticating the cluster autoscaler.

Many components of the Kubernetes API server are hidden in internal packages and not meant to be imported. One example is the controller for service account signing keys.

3

Defaulting webhooks.

Initially, we hoped that we could get away by simply setting the default values using the Helm chart. The controllers disagreed.

4

API server aggregation and CRD race conditions

CRDs are internally managed via the extension server in Kubernetes and not part of the core API. Before starting the controllers, we need to ensure that the CRDs are installed and reconciled.

5

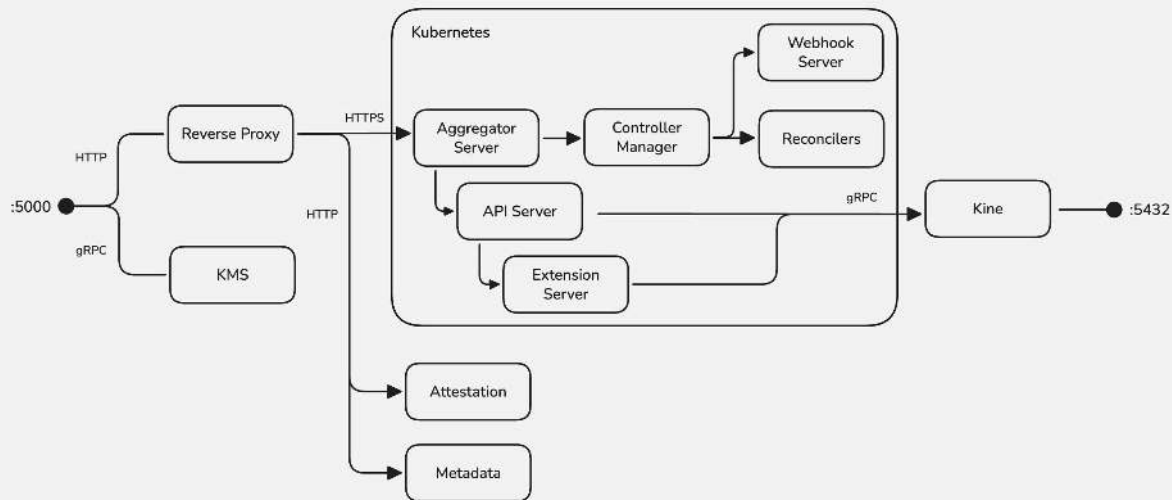
Private network bootstrap

We built a component that allows Talos nodes to automatically bootstrap **etcd** in a private network without requiring an inbound network connection to the Talos API port.

architecture.md

"Simplicity does not precede complexity, but follows it."

- Alan Perlis



\$./kommodity



```
$ grep -r roadmap
```

\$ grep -r roadmap

1

Use the ClusterResourceSet

Currently, we manage the cluster autoscaler via a custom controller. This makes it error-prone to reconcile drift and update the resources associated. The ClusterResourceSet API was built to solve this problem.

2

Simpler image management

Right now, upgrading the Talos version requires some manual preparation of OS images. While minimal and only required for each new Talos version, this is an operational toil that can be automated.

3

Trusted execution environments

We have a few primitives in place to support confidential computing on untrusted infrastructure via attestation, but provider support is spotty and we need to verify different features across providers.

4

Moving away from cloud metadata servers

Part of our security model currently relies on trusting parts of the cloud provider infrastructure. This would not be necessary if we moved away from cloud metadata servers and served metadata via a commodity. To authenticate a machine we would need to verify its integrity first though.

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Help us build the
future of
healthcare.





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